

FoodMart Sales Analytics Project Documentation

Project Overview

The FoodMart Sales Analytics Project is an end-to-end SQL-based business intelligence and analytics solution developed using the FoodMart Sales dataset. The project focuses on transforming raw transactional retail data into meaningful business insights through database design, ETL processes, data quality validation, KPI generation, and analytical reporting.

The solution simulates a real-world retail analytics environment where business stakeholders need visibility into:

- Revenue performance
- Profitability
- Customer behavior
- Product performance
- Regional sales distribution
- Time-based sales trends

The project was implemented using MySQL and demonstrates practical SQL skills used in real data analyst and business intelligence roles.

Business Objective

The primary objective of this project is to help retail decision-makers:

- Monitor overall business performance
 - Identify top-performing products and brands
 - Analyze customer purchasing behavior
 - Detect seasonal sales trends
 - Evaluate regional profitability
 - Improve strategic business decisions using data
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Dataset Overview

Dataset Used: FoodMart Sales Dataset

The dataset contains retail transactional data across:

- Customers
- Products

- Stores
- Regions
- Transactions
- Dates

The project combines transaction data from:

- 1997
 - 1998
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Database Architecture

The project follows a dimensional modeling approach similar to a Star Schema.

Fact Table

Transactions

Contains transactional sales records.

Main Measures:

- Quantity sold
- Revenue
- Profit

Foreign Keys:

- product_id
 - customer_id
 - store_id
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Dimension Tables

Customer

Stores customer demographic and behavioral information.

Examples:

- Gender
- Education
- Income

- Occupation
 - Marital status
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Product

Contains product-related information.

Examples:

- Product name
 - Brand
 - Retail price
 - Cost
 - Product attributes
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Region

Contains sales region and district information.

Date Dimension (dim_date)

Custom-built date dimension used for:

- Time intelligence
 - Monthly analysis
 - Running totals
 - Growth calculations
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ETL Process

The project includes a complete ETL pipeline.

1. Extraction

CSV files were imported into MySQL using:

- LOAD DATA LOCAL INFILE

Data sources included:

- Customer files

- Product files
 - Region files
 - Store files
 - Transaction files
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2. Transformation

Staging Tables

Temporary staging tables were used before loading production tables.

Examples:

- Customer_Staging
- Transactions_Staging

Data Cleaning Performed

- Data type conversion
 - Date formatting using STR_TO_DATE
 - String normalization
 - Numeric casting
 - Removing temporary staging tables after processing
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3. Loading

Cleaned data was inserted into production dimension and fact tables.

Data Quality Validation

The project includes professional data quality checks.

Duplicate Checks

Duplicate validation was performed for:

- Customers
- Products
- Regions
- Stores

- Transactions

Result:

- No duplicate primary keys detected.
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Referential Integrity Checks

Validation ensured:

- Every transaction references a valid product
- Every transaction references a valid customer
- Every transaction references a valid store

Result:

- No orphan records detected.

This confirms high data consistency and integrity.

Core Business KPIs

Total Transactions

269,720 transactions processed.

This indicates:

- Large enough dataset for meaningful analytics
 - Realistic retail transaction simulation
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Total Revenue

\$1,764,546.44

This represents total sales generated during 1997–1998.

Total Profit

\$1,052,818.78

The business maintains a very strong profitability level.

Profit Margin Approximation

$$Profit\ Margin = \frac{1,052,818.78}{1,764,546.44} \times 100 \approx 59.66\%$$

Insight

A profit margin close to 60% indicates:

- Strong pricing strategy
 - Effective cost management
 - High-margin product mix
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Average Order Value (AOV)

AOV = 6.54

Insight

The average transaction size is relatively low, suggesting:

- Customers purchase small quantities frequently
 - Business operates similarly to grocery/FMCG retail patterns
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Time-Based Analysis

Monthly Revenue Trends

The analysis shows strong revenue growth from 1997 to 1998.

Key Observation

Revenue nearly doubled between:

- December 1997 → January 1998

Growth reached:

- +72.31%

This is the highest MoM growth observed.

Seasonal Patterns

Strong Months

Highest-performing months:

- November

- December

Examples:

- Nov 1998 → \$113,787
- Dec 1998 → \$120,160

Insight

Holiday season demand strongly impacts sales.

Weak Months

Consistent dips appear around:

- February
- April
- August
- October

This suggests:

- Seasonal demand fluctuations
 - Possible inventory or promotional gaps
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Running Total Analysis

The running total revenue reached:
\$1.76M by December 1998.

Insight

The cumulative growth curve is consistently upward, indicating:

- Stable business expansion
 - Sustained customer demand
 - Healthy operational performance
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Customer Analytics

Customer Lifetime Value (CLV)

Top customer:

- Ida Rodriguez
- Lifetime Value = \$2235.43

Other high-value customers also contributed significantly.

Insight

A small group of customers contributes disproportionately to revenue.

This supports:

- Loyalty segmentation
 - VIP customer programs
 - Personalized marketing strategies
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Customer Segmentation

The project used:

- NTILE(4) window function

Customers were divided into revenue quartiles.

Business Value

This enables:

- High-value customer targeting
 - Retention campaigns
 - Personalized offers
 - Customer prioritization
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Product Analytics

Top Revenue Products

Highest revenue products include:

- Hermanos Green Pepper
- Hilltop Mint Mouthwash
- Carlson Head Cheese

Insight

The business revenue is distributed across multiple products rather than depending on a single item.

This reduces product concentration risk.

Product Popularity Analysis

Most sold products:

- Tell Tale Fresh Lima Beans
- Steady Whitening Toothpaste
- Hilltop Mint Mouthwash

Important Insight

High sales volume does not always equal high revenue.

Example:

- Toothpaste sold heavily but generated relatively low revenue.

This distinction is important for:

- Pricing strategy
 - Product mix optimization
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Brand Performance Analysis

Top Performing Brands

Leading brands by revenue:

1. Hermanos
 2. Tell Tale
 3. Ebony
 4. Tri-State
 5. High Top
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Brand Insights

Hermanos

- Highest revenue
- Highest profit
- Strong market demand

Tell Tale

- Extremely strong unit sales
- Consistent profitability

Ebony

- Strong balance between volume and profitability
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Long-Tail Brand Distribution

Several brands contribute relatively small revenue.

Examples:

- King
- Toretta
- Symphony

Insight

Some low-performing brands may:

- Require repositioning
 - Need promotional support
 - Be candidates for discontinuation
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Regional Performance Analysis

Top Regions by Revenue

Region	Revenue	Profit
Northwest	847,826	505,783
Mexico Central	330,362	197,072
South West	320,804	191,437

Regional Insights

Northwest Dominance

Northwest contributes nearly half of total revenue.

Insight

This region is the company's primary revenue engine.

Possible reasons:

- Higher customer density
 - Better store performance
 - Stronger demand
 - Better operational efficiency
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Underperforming Regions

Central West

Revenue:

- Only \$9,324

Insight

This region is significantly underperforming.

Potential causes:

- Low customer penetration
 - Weak marketing presence
 - Poor store coverage
 - Limited inventory availability
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Business Recommendations

1. Strengthen High-Performing Regions

Focus on investments:

- Northwest
- Mexico Central
- Southwest

Recommended actions:

- Increasing inventory
- Expand promotions

- Open additional stores
 - Prioritize logistics
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2. Improve Low-Performing Regions

Especially:

- Central West

Recommended actions:

- Investigate operational issues
 - Conduct market analysis
 - Improve regional marketing
 - Reevaluate product assortment
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3. Build Customer Loyalty Programs

Since several customers generate high CLV:

- Introducing VIP rewards
 - Personalized promotions
 - Retention campaigns
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4. Optimize Product Portfolio

Products with:

- High volume but low profitability should be reviewed.

Possible actions:

- Price adjustments
 - Bundling strategies
 - Cost optimization
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5. Prepare for Seasonal Peaks

Strong sales spikes occur in:

- November
- December

Recommendations:

- Increase in stock availability
 - Improve supply chain readiness
 - Launch holiday campaigns early
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6. Focus on High-Profit Brands

Brands such as:

- Hermanos
- Tell Tale
- Ebony

should receive:

- More shelf visibility
 - Larger marketing budgets
 - Strategic expansion
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Technical Skills Demonstrated

This project demonstrates practical expertise in:

SQL Skills

- Joins
 - Aggregations
 - CTEs
 - Window Functions
 - NTILE
 - LAG
 - Running Totals
 - Time Intelligence
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Database Engineering

- Dimensional Modeling
 - Fact & Dimension Design
 - ETL Pipelines
 - Staging Tables
 - Referential Integrity
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Data Analytics

- KPI Development
 - Business Intelligence
 - Customer Analytics
 - Product Analytics
 - Trend Analysis
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Project Conclusion

The FoodMart Sales Analytics Project successfully transformed raw retail transaction data into actionable business intelligence insights.

The analysis revealed:

- Strong overall profitability
- Clear seasonal purchasing patterns
- High-performing brands and regions
- Valuable customer segments
- Opportunities for regional optimization

The project demonstrates the ability to:

- Design scalable analytical databases
- Build ETL workflows
- Perform advanced SQL analytics
- Generate executive-level business insights

This project closely reflects real-world responsibilities in:

- Data Analytics
- Business Intelligence
- Retail Analytics
- SQL Development
- Decision Support Systems